IILM UNIVERSITY, GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

WEEKLY PROGRESS REPORT

Even semester 2026-2027

B.Tech. Major Project Details

Project Id	PT346
Project Title	Autonomous smart irrigation system using ai
Reporting period	15:25. To 17:00
Week No.	6 (08/09/2026)

Work Completed This Week:

So far in this project, we conducted a comprehensive literature review by analyzing thirty research papers on AI-driven and IoT-based smart irrigation, mapping out existing methodologies, evaluation metrics, and key findings. Also we curated over 20 datasets of this topic from sources like kaggle.com and data.gov , and generated more than 10 custom synthetic datasets tailored to our specific modeling requirements.

Challenges/Issues:

In the first week, the biggest challenge was narrowing down a research direction from a huge and scattered pool of smart irrigation papers, most of which were hardware-focused. Filtering relevant, high-quality papers from unreliable sources took considerable time. Finding usable real-world datasets was also difficult, as many were incomplete, region-specific, or not publicly accessible, requiring us to plan synthetic data generation.

Plan for Next Week:

Next, our plan is to finalize the architecture of our software based smart irrigation system. We will decide the input parameters such as soil moisture, weather conditions and crop growth stage then we will study the suitable ai/ml approach. Collect the required dataset and design the initial irrigation decision and prediction model. By the end of the week , we aim to have a basic working prototype.

Guide's Remarks:

Signature of Student

Signature of Guide